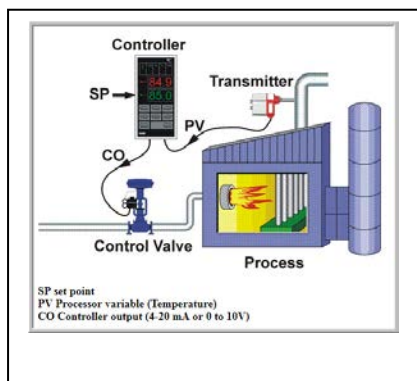


UNIVERSITY OF ILLINOIS AT CHICAGO

Department of Chemical Engineering



College: UIC COLLEGE OF ENGINEERING
Department: UIC Department of Chemical Engineering
Course: CHE 341: Chemical Process Control
Time and Place: SES 238 MW: 09:30 A.M. – 10:45 A.M. F: 10:00 A.M. – 10:50 A.M.
Instructor: Dr. Michael Caracotsios
E-mail: mcaracot@uic.edu
Cell Phone: (630) 696-0934

COURSE OBJECTIVES: This course is designed to extend the student's understanding of the steady-state macroscopic and microscopic material and energy balances into the critical area of process dynamics. The major objective of the course is to emphasize and address the following issues pertinent to the safe and cost effective operation of chemical plants. Given the process and operational targets, the course objectives would be to

design a control system which is (a) stable, (b) has good performance characteristics, and (c) is robust. These design attributes require knowledge of dynamic behavior of processes (process modeling, solution of dynamic equations, characterization of dynamic behavior); control systems; stability and techniques for assessing it; performance criteria and how they are affected by controller parameters. By the end of the course the students should be able to:

- Understand the basic concepts of process dynamic modeling for a variety of chemical engineering systems.
- Know how to develop empirical models from process data using the step response methodology
- Understand the theory of feedback and feed forward control and determine optimal controller settings for stable and robust performance.
- Know when to select and how to implement cascade and feed forward controllers
- Apply available computer software (Matlab, Excel or Athena Visual Studio), or use Laplace transformation theory to solve process dynamic models and investigate their open and closed loop response to various external disturbances.

COURSE PREREQUISITES: The following are key requirements for a successful completion of this course.

- **Integral and Differential Calculus Courses.**
- **Linear First and Second Order Differential Equations.**
- **Familiarity with a Programming Language.**
- **CHE Department Prerequisites: CHE 311, CHE 312, CHE 313, CHE 321 and CHE 396.**

SOFTWARE: Students must download [FREE of charge] from the UIC Blackboard the software programming package [Athena Visual Studio](#) and the EXCEL based Thermodynamic Calculator [CHE301 Calculator.xlsm](#) and the EXCEL based Process Control tool [CHE341 Loop Simulator.xlsm](#)

RECOMMENDED TEXTBOOK AND NOTES:

1. M. Caracotsios, *Chemical Process Control Lecture Notes*, will be distributed free of charge to students, 2020.
2. D.E. Seborg, T.F. Edgar, D.A. Mellichamp, and F.J. Doyle, *Process Dynamics and Control*, Wiley, 2011.

ADDITIONAL REFERENCES:

1. T.E. Marlin, *Process Control: Designing Processes and Control Systems for Dynamic Performance*, McGraw-Hill, 2000.
2. B.A. Ogunnaike and W.H. Ray, *Process Dynamics, Modeling and Control*, Oxford, 1994.
3. J.B. Riggs and M. N. Karim, *Chemical and Bio-Process Control*, Ferret Publishing, Austin, Texas, 2016.
4. G. Stephanopoulos, *Chemical Process Control: An Introduction to Theory and Practice*, Prentice Hall, 1984.

CHE 341: CHEMICAL PROCESS CONTROL COURSE OUTLINE

1. REVIEW AND BACKGROUND

[CHAPTER 1]

- ◆ Introduction to Process Control in Chemical Engineering
- ◆ Elements of Differential and Integral Calculus
- ◆ Numerical Methods for Algebraic and Differential Equations
- ◆ Mass and Energy Balances in the Context of Process Control
- ◆ Tutorials on Programming with VBA and Athena Visual Studio

2. CONTROL OF INDIVIDUAL UNITS

- ◆ Reactors
- ◆ Heat Exchangers
- ◆ Distillation Columns
- ◆ Compressors
- ◆ Other Unit Operations

3. CONTROL CONCEPTS AND DYNAMIC MODELING

[CHAPTERS 3,4,5,6,7]

- ◆ Development of Dynamic Process Models
- ◆ Solution of Dynamic Models and Open Loop Response
- ◆ Laplace Transforms and Transfer Functions
- ◆ Linearization of Process Dynamic Models
- ◆ Process Identification from Plant Data

4. FEEDBACK CONTROL FUNDAMENTALS

[CHAPTERS 8,9]

- ◆ P, PI and PID Controllers
- ◆ Block Diagrams and Routh-Hurwitz Stability Criterion
- ◆ Instrumentation Hardware and Representation

5. FEEDBACK CONTROLLER DESIGN AND TUNING

[CHAPTER 12]

- ◆ Criteria for Controller Tuning
- ◆ PID Controller Tuning Methods
- ◆ Guidelines and Troubleshooting Common Control Loops

6. ADVANCED CONTROL SYSTEMS AND MIMO SYSTEMS

[CHAPTERS 15,16,18,20,22]

- ◆ Feedforward and Cascade Control
- ◆ Interaction and Loop Pairing and Decoupling – Relative Gain Array
- ◆ Control of Common Industrial Process Units
- ◆ Model Predictive Control of Linear Processes
- ◆ Advanced Process Control and Real-Time Optimization

ACADEMIC INTEGRITY:

As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, and administrators—share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the Student Disciplinary Policy:

<https://dos.uic.edu/conductforstudents.shtml>

RELIGIOUS HOLIDAYS:

Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when he/she will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, he/she may request remedy through the campus grievance procedure.

The University Holidays and the Religious Observances calendar can be found online at the following website: <http://oe.uic.edu/religious-calendar/>. The list includes national holidays recognized by the University as well as religious days of special observance that may prohibit a student from attending classes. Please keep in mind this list is not exhaustive.

PARTIAL CREDIT POLICY:

Partial credit can be requested under certain circumstances. The student should make an appointment with the TA/Professor and be prepared to discuss the reasons for requesting the extra credit. Final decision rests with the Professor.

Partial credit can also be requested in cases where there is an obvious (indisputable) error made by the Professor or the Teaching Assistant.

No partial credit will be given for:

1. Incorrect formulas, full or partial.
2. Inconsistent or missing units in calculations.
3. Multiple answers.
4. Incoherent (illogical, disjointed) derivations or calculations.

STUDENT COURTESY POLICY

Talking/Smart Phones/Laptops/Food and Drinks in Class:

Not allowed unless explicitly stated

Repeat Offenders will be asked to leave the classroom

Cheating/Copying: The entire class gets penalized after the first warning

Reduce the homework contribution to the final grade

Perpetrators will receive zero points in their homework

Late Homework and Disputes:

Late Homework will not be accepted without prior arrangements with your Professor in person

If there are Homework and Exam score disputes they will be resolved only with the TA/Professor.